Bihar Engineering University, Patna

B.Tech. 8th Semester Examination, 2024

Course: B.Tech. *Time: 03 Hours*
Code: 105819 *Subject: Ad-hoc and Sensor Networks* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct option / write the answer of the following (any seven only):*** **[2 x 7 = 14]**

(a) Ad-hoc Wireless Networks are:
(i) Infrastructure less network
(ii) Fixed Infrastructure less network
(iii) Both (i) and (ii)
(iv) None of the above

(b) Differentiate between WSNs and MANETs.

(c) How CSMA based MAC work on Wireless Sensor Network?

(d) Write down the difference between Proactive routing and Reactive routing.

(e) What is data aggregation strategy in Wireless Sensor Network?

(f) List out the hardware and software components of a sensor node.

(g) What is hybrid routing?

(h) List any three radio waves propagation mechanism.

(i) Write the classification of MAC protocols?

(j) List the issues in designing a transport layer protocol for ad hoc wireless networks.

Q.2 (a) Describe the characteristics, requirements and applications of Ad-hoc and Sensors Networks. **[7]**

(b) Explain briefly the architecture of MANET with a neat diagram. **[7]**

Q.3 (a) Explain in detail the principle of contention-based protocols with scheduling mechanism. **[7]**

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(b) List and explain the issues and goals in designing a MAC protocol for ad-hoc wireless networks. **[7]**

Q.4 (a) What is TCP? Discuss TCP over Ad-hoc Wireless Networks with an example. **[7]**

(b) Discuss any two Reactive routing protocols for ad-hoc wireless networks. **[7]**

Q.5 (a) Explain in detail about the single node architecture in wireless networks. **[5]**

(b) How hybrid TDMA/FDMA medium access control protocol for wireless sensor networks work? Explain with an example. **[9]**

Q.6 (a) Write in detail about the IEEE 802.15.4. **[7]**

(b) What is a wireless sensor network? Explain the wireless sensor network architecture with a diagrammatic illustration. **[7]**

Q.7 (a) Describe in detail the design issues in routing and transport layer protocols. **[7]**

(b) Discuss the Characteristics of the wireless channel. **[7]**

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Q.8 (a) Explain water marking techniques in details. [7]

(b) Identify the Key management schemes in Wireless Sensor Networks. Illustrate the key distribution and management mechanism required for secure communication in sensor networks. [7]

Q.9 Write Short notes on ***any two*** of the following [7 x 2=14]

(a) Intrusion detection

(b) MAC protocol

(c) Data relaying strategies

(d) IEEE 802.11

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